

Principles of Programming Languages

Lecture 2: Inductive definitions (in Coq).

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Outline

Inference rules

Derivations

Proofs and proofs by induction

Inference rules

- ▶ Inference rules = *rule schemes*
- ▶ Example: natural numbers

$$\frac{\cdot}{0 \in \mathbb{N}} \text{ zero}$$

$$\frac{n \in \mathbb{N}}{S n \in \mathbb{N}} \text{ succ}$$

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- ▶ We will use them to define the syntax and the semantics of programming languages

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Naturals in Coq

- ▶ The syntax of natural numbers:

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- ▶ In Coq:

```
Inductive Nat := O : Nat  
           | S : Nat -> Nat.
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Inference rules

- Definition:

$$\frac{\cdot}{0 \in \mathbb{N}} \text{ zero}$$

$$\frac{n \in \mathbb{N}}{S\ n \in \mathbb{N}} \text{ succ}$$

- Derivation example for $S\ (S\ 0) \in \mathbb{N}$:

$$\frac{\frac{\frac{\cdot}{0 \in \mathbb{N}} \text{ zero}}{S\ 0 \in \mathbb{N}} \text{ succ}}{S\ (S\ 0) \in \mathbb{N}} \text{ succ}$$

- In Coq:

```
Check S (S 0) .  
S (S 0)  
: Nat
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Semantics of naturals in Coq

- Recall syntax:

$$\frac{\cdot}{0 \in \mathbb{N}} \text{ zero}$$

$$\frac{n \in \mathbb{N}}{S\ n \in \mathbb{N}} \text{ succ}$$

- Addition:

$$0 + m = m$$

$$S\ n + m = S\ (n + m)$$

- In Coq:

```
Fixpoint plus (n m : Nat) : nat :=  
  match n with  
  | 0 => m  
  | S n' => S (plus n' m)  
end.
```

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Evaluation

```
Eval compute in (plus 0 0).
```

```
= 0
```

```
: Nat
```

```
Eval compute in (plus 0 (S 0)).
```

```
= S 0
```

```
: Nat
```

```
Eval compute in (plus (S 0) 0).
```

```
= S 0
```

```
: Nat
```

```
Eval compute in (plus (S 0) (S 0)).
```

```
= S (S 0)
```

```
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```

```
Eval compute in (plus (S (S 0)) (S (S 0))).
```

```
= S (S (S (S 0)))
```

```
: Nat
```

Proof by simplification

Lemma plus_0_m_is_m :
 forall m, plus 0 m = m.

Proof.

(first, we move m from the quantifier
 in the goal to a context of current
 assumptions using the 'intro' tactic *)*

intros m.

(the current goal is now provable
 using the definition of plus: we
 apply the 'simpl' tactic for that*)*

simpl.

(the current goal is just reflexivity *)*

reflexivity.

(Qed checks if the proof is correct *)*

Qed.

Induction principles

- ▶ Recall the inductive definition for $\mathbb{N}$:

$$\frac{\cdot}{0 \in \mathbb{N}} \text{ zero}$$

$$\frac{n \in \mathbb{N}}{S n \in \mathbb{N}} \text{ succ}$$

- ▶ What is a proof by induction in this case?
- ▶ Suppose that we want to prove $P(n)$ where $n \in \mathbb{N}$ and P is a property
- ▶ $P(n)$ holds if:
 - ▶ $P(0)$ holds - this corresponds to *zero*
 - ▶ $P(n)$ implies $P(n+1)$ - which corresponds to *succ*

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Induction principles

- For naturals defined as below $\mathbb{N}$:

$$\frac{\cdot}{0 \in \mathbb{N}} \text{ zero}$$

$$\frac{n \in \mathbb{N}}{S n \in \mathbb{N}} \text{ succ}$$

there is a corresponding induction principle:

$$\forall P. (P(0) \wedge (\forall n. P(n) \rightarrow P(S n))) \rightarrow \forall n. P(n)$$

- Proofs by induction are based on induction principles

Induction principles in Coq

- ▶ In Coq induction principles are generated automatically
- ▶ Recall the Coq definition of naturals:

```
Inductive Nat := O : Nat
              | S : Nat -> Nat.
```

- ▶ When the definition is processed, Coq generates (among others) the `Nat_ind` induction principle (demo)
- ▶ **Check** `Nat_ind`.

```
Nat_ind
: forall P : Nat -> Prop,
  P O ->
  (forall n : Nat, P n -> P (S n)) ->
  forall n : Nat, P n
```

Proofs by induction: example 1

- ▶ Lemma: $P = \forall n. n + 0 = n$
- ▶ Can we prove this by applying the definitions?

$$\frac{\cdot}{0 \in \mathbb{N}} \text{ zero}$$

$$\frac{n \in \mathbb{N}}{S\ n \in \mathbb{N}} \text{ succ}$$

$$0 + m = m \text{ (base)}$$

$$S\ n + m = S\ (n + m) \text{ (ind)}$$

- ▶ Induction principle:
 $\forall P. (P(0) \wedge (\forall n. P(n) \rightarrow P(S\ n))) \rightarrow \forall n. P(n)$
- ▶ Proof by induction

1. case $P(0)$: $0 + 0 = 0$, by (base)
2. case $n = S\ n'$:

- ▶ Inductive hypothesis (IH): $P(n)$: $n + 0 = n$
- ▶ Prove $P(S\ n)$: $S\ n + 0 =$ (by ind) $S\ (n + 0) =$ (by IH) $S\ n$

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Example 1 in Coq

```
Lemma plus_n_0_is_n :  
  forall n,  
    plus n 0 = n.
```

```
Proof.
```

```
(* Exercise: why this does not work by simplification? *)  
(* Induction by n using Nat_ind principle:  
    note that two goals are generated. *)
```

```
induction n.
```

```
- (* solve the first goal by simplification *)
```

```
  simpl.
```

```
  reflexivity.
```

```
- (* simplifies just a part of the goal *)
```

```
  simpl.
```

```
(* apply the inductive hypothesis by  
    rewriting conveniently the expression *)
```

```
  rewrite IHn.
```

```
  reflexivity.
```

```
Qed.
```

Example 2 in Coq

```
Lemma plus_comm :  
  forall n m, plus n m = plus m n.  
(* DEMO *)
```

Demo: lists in Coq

- ▶ Defining lists in Coq
- ▶ Induction principle
- ▶ Functions: append, reverse
- ▶ Proofs: append nil left/right, associative append, involutive reverse

Demo: trees in Coq

- ▶ Defining trees in Coq
- ▶ Induction principle
- ▶ Functions: search, preorder
- ▶ Proofs: search is correct

Where do we need trees?

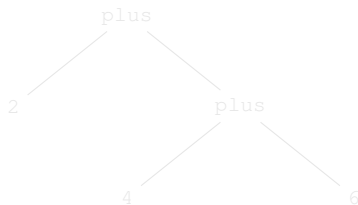
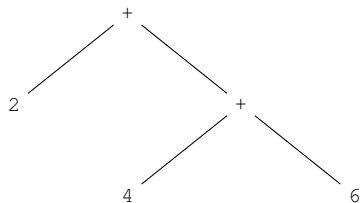
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- ▶ An AST is a *tree* representation of the structure of a program
- ▶ "Abstract" = not every detail occurring in the text of the program is present in the AST representation
- ▶ Compilers use ASTs as the main data structure

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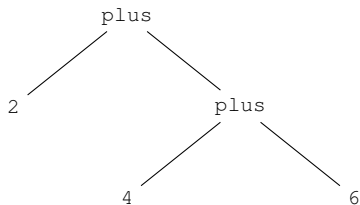
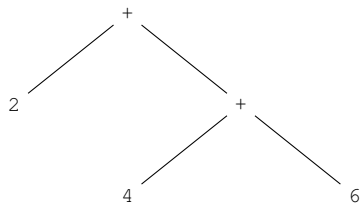
Example: arithmetic expressions

AST for $2 + (4 + 6)$:



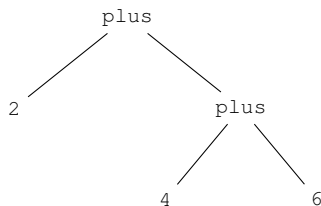
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```
Inductive Exp :=  
| number : nat -> Exp  
| plus : Exp -> Exp -> Exp.
```

```
Coercion number : nat >-> Exp.
```

```
Check (plus 2 (plus 4 6)).  
plus 2 (plus 4 6)  
: Exp
```

Conclusions

► Intensively used concepts:

1. inductive definitions
2. induction principles
3. functions
4. proofs

► Bibliography:

1. Chapter Inductive Definitions in Practical Foundations of Programming Languages, Robert Harper
<https://www.cs.cmu.edu/~rwh/pfpl/2nded.pdf>
2. Chapter Proof by induction in Software Foundations - Volume 1, Benjamin C. Pierce, Arthur Azevedo de Amorim, Chris Casinghino, Marco Gaboardi, Michael Greenberg, Cătălin Hrițcu, Vilhelm Sjöberg, Andrew Tolmach, Brent Yorgey
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